Phrase 8 and 9

Table 1. Frequency CNN Classification Report

Class	Precision	Recall	F1-Score	Support
REAL	0.82	0.86	0.84	10,000
FAKE	0.85	0.81	0.83	10,000
Accuracy	—	—	**0.83**	**20,000**
Macro Average	0.83	0.83	0.83	20,000
Weighted Average	0.83	0.83	0.83	20,000

Table 2. Spatial CNN vs. Frequency CNN Performance Comparison

Metric	Spatial CNN	Frequency CNN	Improvement (Δ)
Accuracy	0.5065	0.8326	+0.3261
Precision	0.5921	0.8507	+0.2586
Recall	0.0418	0.8068	+0.7650
F1-Score	0.0781	0.8282	+0.7501
Specificity	0.9712	0.8584	−0.1128
AUC-ROC	0.4609	0.9140	+0.4530

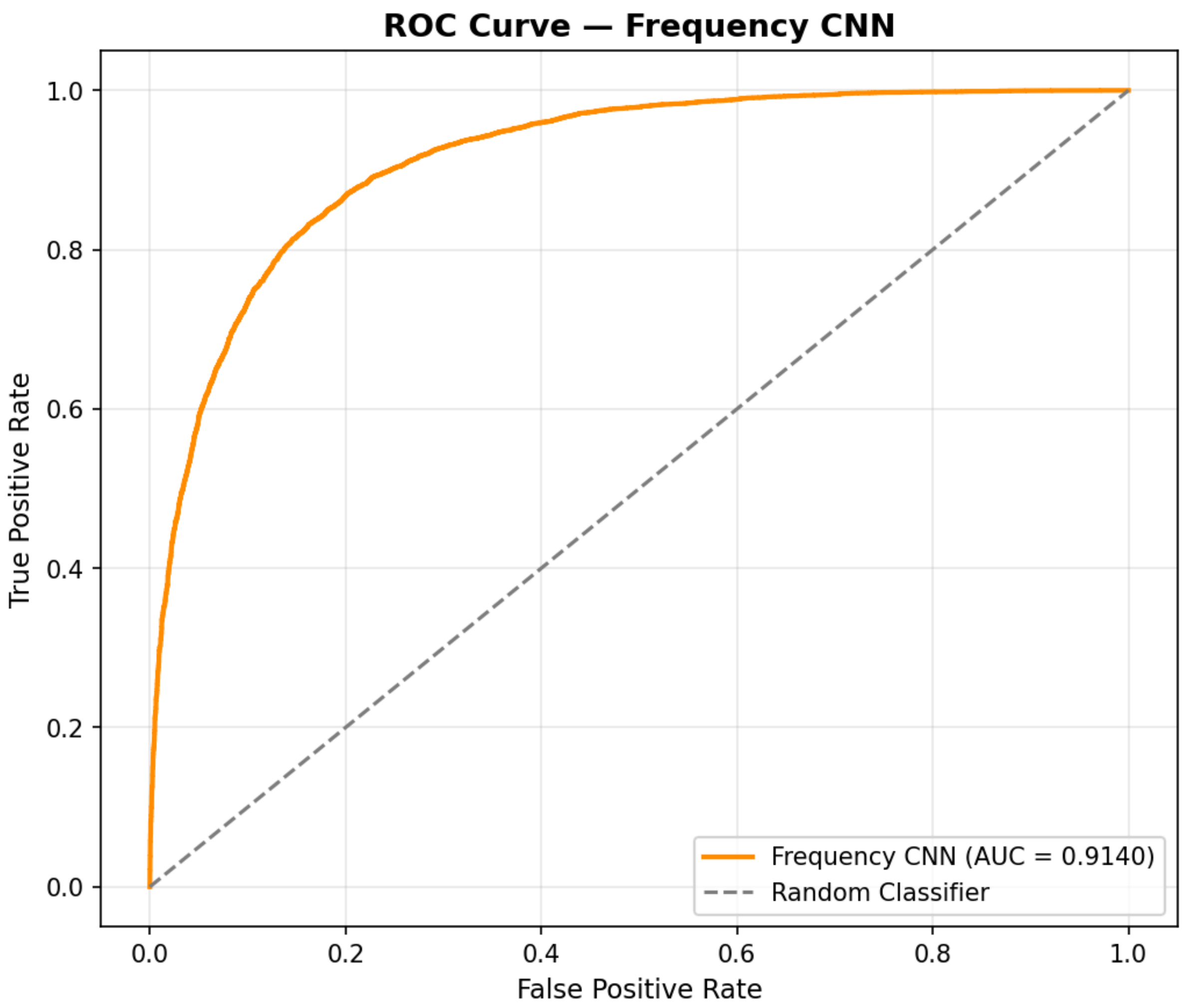
Table 3. Baseline Spatial CNN Performance (Phase 3 Reference)

Metric	Value
Accuracy	0.6826

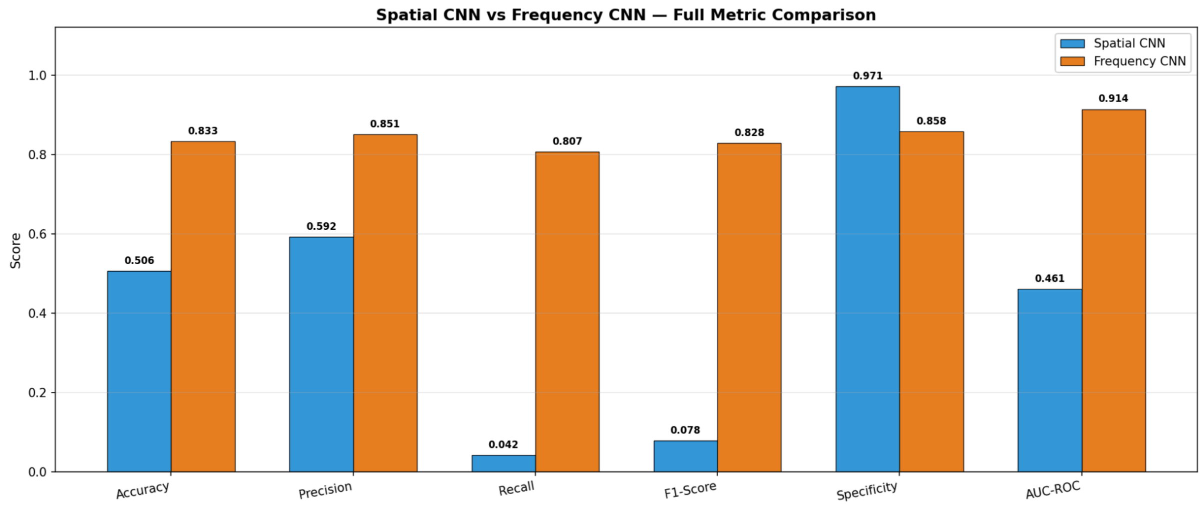
Precision	0.6434
Recall	0.8194
F1-Score	0.7208
Specificity	0.5458
AUC-ROC	0.7626

Table 4. Performance Interpretation

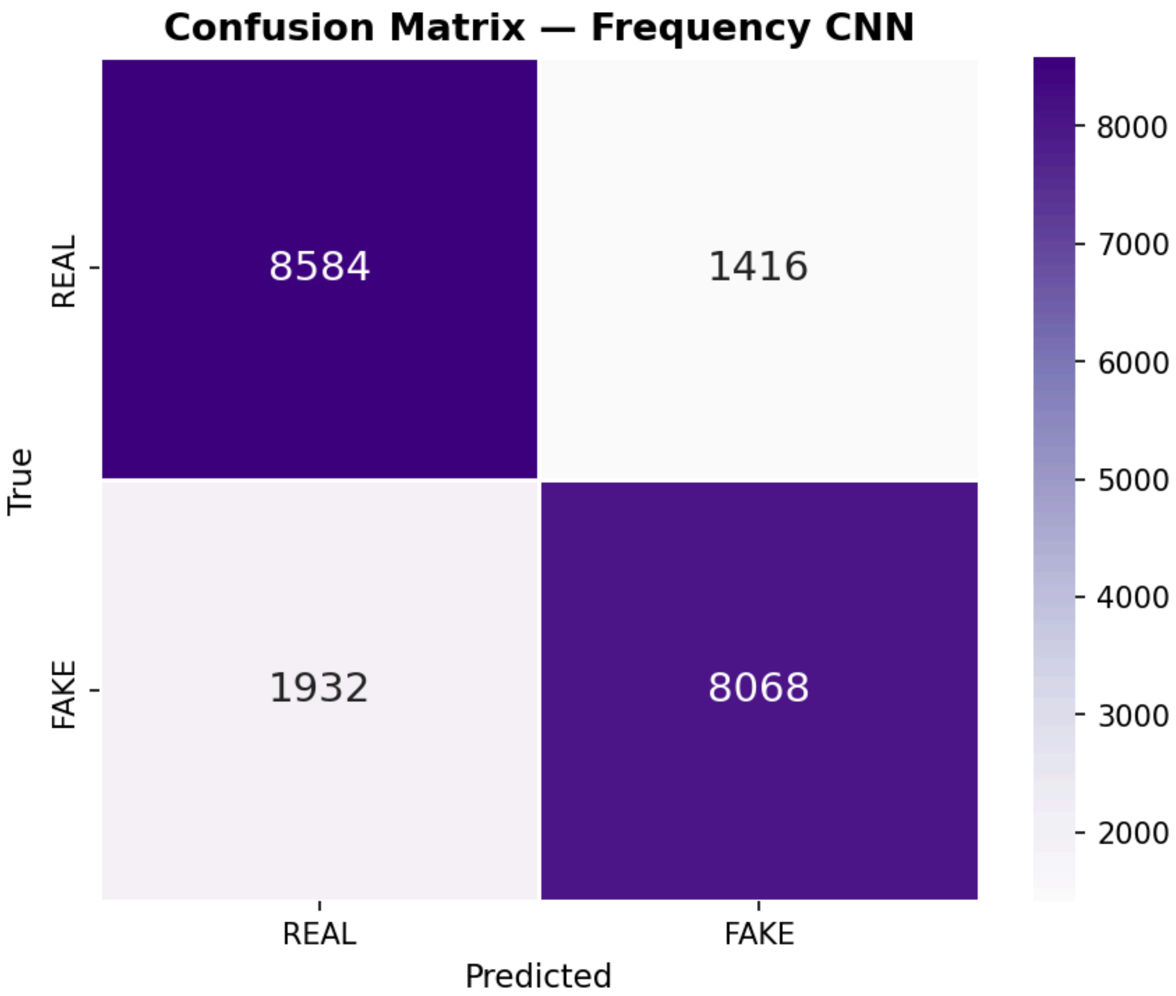
Observation	**Interpretation**
Accuracy	Frequency CNN improves overall classification accuracy from 50.65% to 83.26%.
Recall	Recall increases substantially, indicating much better detection of fake images.
F1-Score	The large increase in F1-score demonstrates a better balance between precision and recall.
AUC-ROC	AUC-ROC improves from 0.4609 to 0.9140, showing significantly stronger class separability.
Specificity	Although specificity decreases slightly, the overall detection performance is greatly improved.
Key Finding	Frequency-domain features provide substantially richer discriminative information than spatial-only features for deepfake detection.

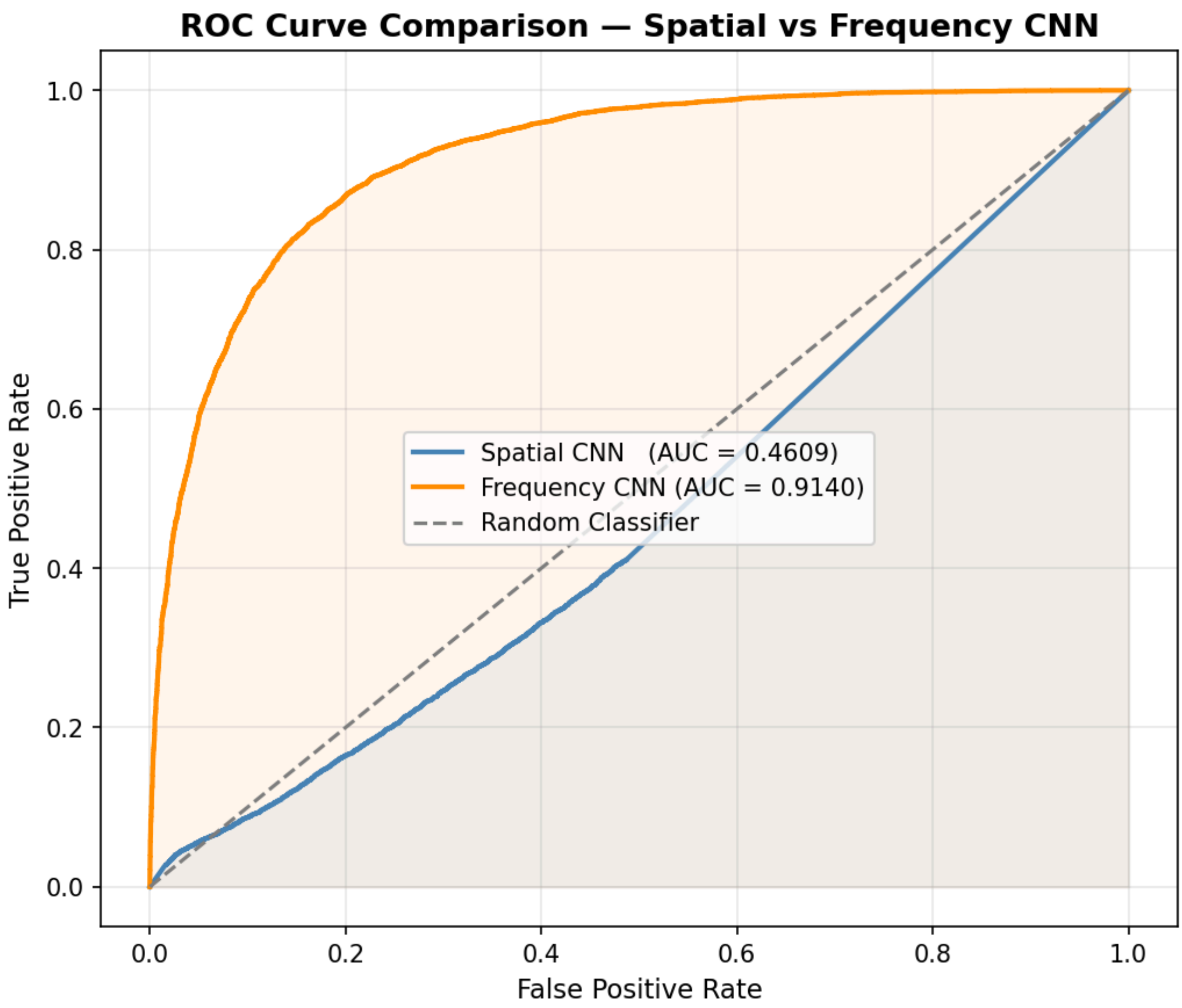


ROC Curve — Frequency CNN
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Frequency CNN (AUC = 0.9140)
Random Classifier



Spatial CNN vs Frequency CNN — Full Metric Comparison
Score
Spatial CNN
Frequency CNN
0.506
0.833
0.592
0.851
0.042
0.807
0.078
0.828
0.971
0.858
0.461
0.914
Accuracy
Precision
Recall
F1-Score
Specificity
AUC-ROC


Confusion Matrix — Frequency CNN
True
REAL
FAKE
8584
1416
1932
8068
Predicted
8000
7000
6000
5000
4000
3000
2000


ROC Curve Comparison — Spatial vs Frequency CNN
Spatial CNN (AUC = 0.4609)
Frequency CNN (AUC = 0.9140)
Random Classifier
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0